

Load Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1797

Virginia Tech Power Plant 138 kV
August 16, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1797. The Interconnection Customer proposes a 182.4 MW large load (data center) facility interconnecting at the Virginia Tech Power Plant 138 kV substation in the Virginia area. The request was received on June 25, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 592 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$4,731,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Virginia Tech Power Plant / Hartman Creek 138 KV Line	Shullsburg / Edgemont 138 KV	118.1	282	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Virginia Tech Power Plant / Hartman Crk 138 KV	Rebuild line with higher-capacity conductor	\$4,731,000	100% (\$4,731,000)
	Total allocated to project		\$4,731,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.